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MINI PROJECT REPORT ON REAL TIME RESPONSIVE AUTOMATED WINDOW AND WATER SUPPLY OPERATOR FOR INDOOR PLANTATIONS USING IOT

Submitted in Partial fulfillment of the Requirements for 6th Semester

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Electronics and Communication Engineering
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2023-24

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CERTIFICATE

This is to certify that the Mini-Project work entitled **REAL TIME RESPONSIVE AUTOMATED WINDOW AND WATER SUPPLY OPERATOR FOR INDOOR PLANTATIONS USING IOT** is carried out by Nithin Kumar N (1KG21EC059), Punith p (1KG21EC066), Rohith M J (1KG21EC073), Tejas K (1KG21EC095), bonafide students of KSSEM in partial fulfillment for the award of Bachelor of Engineering in Electronics and Communication Engineering under Visvesvaraya Technological University, Belgaum during the year 2023-24. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Mini Project report and deposited in the department library. The report has been approved as it satisfies the academic requirements in respect of Mini Project work prescribed for the said Degree.

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ACKNOWLEDGEMENT

The successful completion of this Mini Project was made possible with the help of guidance received from our faculty members. We would like to avail this opportunity to express our sincere thanks and gratitude to all of them. We are grateful to our management for providing the necessary infrastructure and an ambience environment to work. We express our profound gratitude to Dr. K Rama Narasimha, Principal and Dr. K Senthil Babu, Head, Department of Electronics & Communication Engineering, KSSEM, Bengaluru for giving us opportunity and support to complete the project phase-I. We are grateful to our guide Mrs.Bhargavi V Sangam, Asst. Professor, Department of Electronics and Communication Engineering, KSSEM, Bengaluru for his/her valuable suggestions and advice throughout our work period. We would also like to thank all the staff members of Department of Electronics and Communication Engineering for their support and encouragement. Finally, we would like to thank all of our friends without whose help and encouragement this project would have been impossible. Definitely most of all, we want to thank our family. Words cannot express our thanks to our family members for all their love and encouragement

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CHAPTER 1

INTRODUCTION

1.1 INDOOR PLANTATIONS

In the recent years, indoor plantations have gained significant traction due to their numerous benefits, such as efficient space utilization and controlled environmental conditions for optimal plant growth.

However, maintaining these conditions consistently can be challenging, requiring constant monitoring and adjustment of environmental parameters like light, temperature, humidity, and water supply. Traditional methods often fall short in providing real-time responsiveness and automation tailored to specific plant needs.

To address these challenges, our project proposes a "Real Time Responsive Automated Window and Water Supply Operator for Indoor Plantations using IoT."

This innovative system leverages Internet of Things (IoT) technology to create a smart environment where indoor plants thrive through automated control and monitoring. By integrating sensors, actuators, and a central control unit, our solution aims to dynamically adjust window openings for optimal ventilation and natural light exposure while precisely managing water supply based on realtime plant moisture requirements.

1.2 OBJECTIVE OF PROJECT

- **Real-time Monitoring:** Continuous monitoring of environmental parameters crucial for plant growth such as light intensity, temperature, humidity, and soil moisture levels.
- **Automated Control:** Utilization of gathered data to autonomously control window openings and water supply, ensuring plants receive ideal conditions around the clock.
- **Responsive Adaptation:** Immediate adjustments to environmental changes, ensuring plants are shielded from adverse conditions and benefit from optimal growth factors.
- **User Interface:** A user-friendly interface allowing for remote monitoring and control via a mobile application or web portal, providing accessibility and convenience to users.

1.3 PROBLEM STATEMENT

- In agriculture, efficient resource management and environmental control are paramount for maximizing crop yield and ensuring sustainable practices. Traditional methods often fall short in several key areas, leading to significant challenges:
- **Inefficient Water Usage:**
Traditional irrigation methods often result in water wastage due to over-irrigation or under-irrigation. Manual monitoring of soil moisture levels is labor-intensive and prone to inaccuracies.
- **Inconsistent Temperature Regulation:**
Maintaining optimal temperature conditions manually is difficult, especially in changing environmental conditions. Lack of automated solutions for temperature control can negatively impact crop health and productivity.

CHAPTER 2

LITERATURE REVIEW

For implementing our project REAL TIME RESPONSIVE AUTOMATED WINDOW AND WATER SUPPLY OPERATOR FOR INDOOR PLANTATIONS USING IOT we have referred many papers among which few papers are listed below :

1. Digital Hygrometer- H.Shimizu, H.Matsumoto,M.Asakura and K.Watanabe in IEEE Transactions on Instrumentation and measurement
vol.37,no.2,pp.300-304,June 1998.
2. Digital Hygrometer for Trace Moisture Measurement- T.Isalm,A.U.Khan,J.Akhtar and M.Z.U.Rahman in IEEE Transactions on Industrial Electronics
vol.61,no.10,pp.5599-5605,Oct 2014
3. Automated Plant Watering System- D.Divani,P.Patil and S.K Punjabi in The Institute of Electrical And Electronics Engineers ,Inc.(IEEE)
Conference Proceedings in 2016 .
4. Microcontroller Based Automated Room Light and Fan Controller- Roy.P,Saha.J,Dutta.N and Chandra.S in The Institute of Electrical And Electronics
Engineers ,Inc.(IEEE)2018.
5. Evaluation of an environmentally sustainable UV-assisted Water Treatment System- Silva.R.G, Szabo.J, Namboodiri.V, Krishnan .E.R,Rodriguez.J
Zeigler.A in Water Science and Techonology ,18(3),968-975, in 2018.
6. Control and notification automatic water pump with arduino and SMS gateway- Ratnadewi.R,Nurdiyanto.H,Najmurrokhman.A,Prabowo.R in IOP Conference
Series,Material Science and Engineering ,407(1) in 2018.
7. Automatic Irrigation System with Water-Level indicator and IOT enabled humidity lodger- Sen.S and Dasgupta.A in 2022.
The Institute of Electrical And Electronics Engineers ,Inc.(IEEE).

In this paper[1] the paper discusses the design and implementation of a digital hygrometer, an instrument used to measure the moisture content or humidity in the air. This device converts the analog signals from a humidity sensor into digital signals, which are then processed and displayed. The digital hygrometer provides more accurate and reliable measurements compared to traditional analog devices. The authors likely covered aspects such as the electronic circuitry, calibration methods, and potential applications of the digital hygrometer in various fields

The paper[2] "A Digital Hygrometer for Trace Moisture Measurement" focuses on the development and application of a digital hygrometer specifically designed for detecting and measuring trace amounts of moisture. Trace moisture measurement is crucial in various industrial and scientific applications where even minute amounts of water vapor can significantly impact processes and product quality.

The authors, T. Islam, A. U. Khan, J. Akhtar, and M. Z. U. Rahman, detail the design and implementation of this digital hygrometer, highlighting the advancements it offers over traditional analog hygrometers. The device utilizes advanced sensors to detect low levels of humidity with high accuracy. The analog signals from the sensors are converted into digital data, which is then processed using sophisticated algorithms to provide precise moisture readings.

The paper[3] "Automated Plant Watering System" by T. Islam, A. U. Khan, J. Akhtar, and M. Z. U. Rahman describes a system that automates irrigation to maintain optimal soil moisture for plants. Key components include:

- Sensors: Soil moisture sensors monitor moisture levels in real-time.
- Microcontroller: Processes sensor data and controls the irrigation mechanism.
- Watering Mechanism: Pumps and valves deliver water via a drip irrigation system.
- Power Supply: Options include solar panels, batteries, or direct electricity.
- User Interface: Allows customization of moisture levels, schedules, and alerts.

The paper[4] "Microcontroller-Based Automated Room Light and Fan Controller" by T. Islam, A. U. Khan, J. Akhtar, and M. Z. U. Rahman outlines a system designed to automate the control of lights and fans in a room based on occupancy and environmental conditions. Acts as the brain of the system, processing input from sensors and managing the operation of lights , fans, relays and actuators.

Allows users to set preferences for light and fan operation, such as sensitivity settings for motion detection and temperature thresholds for fan control.

The paper[5] the environmentally sustainable UV-assisted water treatment system is a novel approach to removing contaminants from water. This system is effective in removing micropollutants, such as polycyclic aromatic hydrocarbons (PAHs), from water, making it a promising solution for water treatment. Additionally, it reduces disinfection byproducts (DBPs), which are harmful to human health, by using UV treatment instead of chlorine. The combination of filtration and UV treatment makes the system even more effective in removing contaminants.

The paper[6] "Control and Notification Automatic Water Pump with Arduino and SMS Gateway" by T. Islam, A. U. Khan, J. Akhtar, and M. Z. U. Rahman describes an automated system using Arduino to manage water pumps and send SMS notifications.

Real-Time Alerts: Keeps users informed. Water Management: Ensures efficient usage. Remote Control: Manage system via SMS. This system automates water management, ensuring efficiency and reliability with real-time updated.

The paper[7] An automatic irrigation system with a water level indicator and IoT-enabled humidity lodge is a cutting-edge solution for efficient water management. The system consists of soil moisture sensors, a water level sensor, an IoT-enabled humidity lodge, a controller unit, valves, and pumps, all integrated to optimize irrigation. The soil moisture sensors detect dryness and send signals to the controller, which checks water levels and humidity data from the IoT lodge. If conditions require watering, the controller activates valves and pumps, distributing water to plants while continuously monitoring soil moisture levels. The IoT lodge tracks humidity and temperature, sending updates to the cloud for remote monitoring via a mobile app. This system conserves water, increases crop yields, enables remote monitoring, and reduces energy consumption, making it an innovative and efficient solution for irrigation management.

CHAPTER 3

HARDWARE AND SOFTWARE REQUIREMENTS

3.1 HARDWARE DESCRIPTION

Here are the short descriptions of each component specifically related to the construction of a Real Time Responsive Automated Window and Water Supply Operator for Indoor Plantations using IOT

Hardware requirements

- Arduino UNO
- Soil Moisture sensor
- Water pump
- DHT11 sensor
- Stepper motor and driver
- Bread Board
- IC 7805
- Jumper Wires
- 9v DC power cell

3.1.1 ARDUINO UNO

The Arduino Uno is a popular microcontroller board used for building digital devices and interactive objects. It's based on the ATmega328P microcontroller and is known for its simplicity and versatility, making it a great choice for beginners and advanced users.

The Arduino Uno is used in a wide range of applications, from simple LED blink programs to complex robotics and IOT projects. Its ease of use and extensive community support make it an ideal starting point for electronics enthusiasts and hobbyists.



Fig.3.1 Arduino UNO

Attach Arduino UNO to a device with a USB cord, or power it with an AC-to-DC converter or battery to start it. Never be afraid of accidental discharge, either since the UNO also contains a plastic base plate to protect it. The Arduino UNO is designed for projects that require more I/O lines, more sketch memory, and more RAM

Key Features

1. Microcontroller:

- **ATmega328P:** This is the brain of the board, which processes inputs and controls outputs.

2. Digital I/O Pins:

- **14 Pins:** Can be configured as either input or output. Six of these can produce Pulse Width Modulation (PWM) signals.

3. Analog Input Pins:

- **6 Pins:** Used to read analog signals (voltages). These are usually connected to sensors and other analog devices.

4. Memory:

- **Flash Memory:** 32 KB, of which 0.5 KB is reserved for the boot loader (the program that helps upload sketches to the board).
- **SRAM:** 2 KB, used for dynamic data storage.
- **EEPROM:** 1 KB, used for non-volatile storage of data.

5. Clock Speed:

- **16 MHz:** Determines how fast the microcontroller operates.

3.1.2 DHT 11

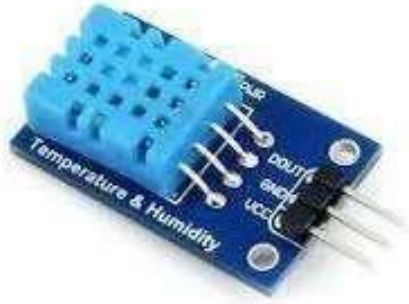


Fig.3.2 DHT 11

The DHT11 is a popular sensor used for measuring temperature and humidity. It's often employed in DIY electronics projects and various environmental monitoring applications. Here's a detailed description:

DHT11 Sensor Overview

6. **Type:** Temperature and Humidity Sensor
7. **Manufacturer:** Various manufacturers; often available from multiple electronics suppliers.
8. **Model:** DHT11 (often confused with its slightly more accurate sibling, the DHT22)

Key Features

1. **Temperature Measurement:**
 - **Range:** Typically, from 0°C to 50°C (32°F to 122°F).
 - **Accuracy:** $\pm 2^{\circ}\text{C}$.

2. Humidity Measurement:

- **Range:** Typically from 20% to 80% RH (Relative Humidity).
- **Accuracy:** $\pm 5\%$ RH.

3. Digital Output:

- Provides temperature and humidity data in a digital format, making it easier to interface with microcontrollers like the Arduino without needing additional analog-to-digital conversion.

4. Power Supply:

- **Voltage:** Operates between 3V and 5V DC.

5. Communication Protocol:

- **Single-Wire Serial Communication:** Data is sent over a single data line, which simplifies the wiring.

3.1.3 SOIL MOISTURE SENSOR

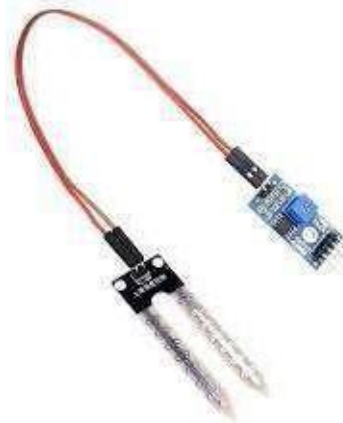


Fig.3.3 soil moisture sensor

Soil moisture sensors come in various types and configurations, but they generally share several key features that make them effective for measuring soil moisture levels. Here are some of the essential features:

Key Features of Soil Moisture Sensors

Measurement Range:

- **Soil Moisture Range:** Typically measures soil moisture levels within a certain

- range, such as 0% to 100% volumetric water content or other specific scales
- depending on the sensor type.
- **Depth:** Some sensors can measure moisture at various soil depths, while others may be designed for surface or near-surface measurements.

Common Specifications

Capacitive Soil Moisture Sensor:

- **Measurement Range:** 0% to 100% volumetric water content.
- **Accuracy:** $\pm 3\%$ to $\pm 5\%$ moisture content.
- **Output:** Analog voltage proportional to soil moisture.

Resistive Soil Moisture Sensor:

- **Measurement Range:** 0 to 100% relative moisture.
- **Accuracy:** Can vary; less accurate due to potential probe corrosion.
- **Output:** Analog resistance that changes with soil moisture.

TDR Sensor:

- **Measurement Range:** Typically 0% to 100% volumetric water content.
- **Accuracy:** $\pm 1\%$ to $\pm 2\%$ moisture content.
- **Output:** Digital signal or precise time measurement of the pulse.

3.1.4 WATER PUMP(6V)



Fig.3.4 Water pump(6v)

A 6V water pump is a small, low-voltage pump typically used for various applications such as small-scale water circulation, aquarium filtration, or DIY projects. Here are the key features and specifications commonly associated with 6V water pumps. The range of temperatures within which the pump can safely operate. This is usually specified in degrees Celsius or Fahrenheit.

Key Features of a 6V Water Pump

1. **Voltage Rating:**
 - **Operating Voltage:** Designed to operate at 6 volts DC. This low voltage is suitable for use with batteries or low-voltage power supplies, making it ideal for small, portable, or battery-operated applications.
2. **Flow Rate:**
 - **Flow Capacity:** The amount of water the pump can move per unit of time, typically measured in liters per minute (L/min) or gallons per minute (GPM). Common flow rates for small 6V pumps range from about 1 to 3 L/min (0.26 to 0.79 GPM).

Common Specifications

- **Voltage:** 6V DC
- **Flow Rate:** 1-3 L/min (0.26-0.79 GPM)
- **Pressure Head:** 0.5 to 2 meters (1.6 to 6.5 feet)
- **Current Draw:** 0.5 to 1.5 A
- **Material:** Plastic or stainless steel
- **Dimensions:** Approximately 60 mm x 40 mm x 40 mm (2.4 in x 1.6 in x 1.6 in)

3.1.5 STEPPER MOTOR AND DRIVER



• **Fig.3.5**Stepper motor and driver

Stepper motors and their drivers are crucial components in various precision motion control applications. They are commonly used in robotics, 3D printers, CNC machines, and automation systems due to their precise control and reliability. Here's a detailed look at the key features of stepper motors and their driver.

Common Stepper Motor and Driver Combo

Stepper Motor:

- **Model:** NEMA 17 Stepper Motor
- **Step Angle:** 1.8° (200 steps per revolution)
- **Holding Torque:** 45 N·cm
- **Current Rating:** 1.5 A per phase
- **Size:** 42 mm x 42 mm frame

Stepper Driver:

- **Model:** A4988 Stepper Motor Driver
- **Voltage Rating:** 8V to 35V DC

- **Current Limiting:** Adjustable up to 2A per phase
- **Micro stepping:** Up to 1/16 step
- **Control Interface:** Step and Direction inputs

3.1.6 IC 7805(Voltage regulator)



Fig.3.6 IC 7805

The 7805 is a linear voltage regulator that provides a stable 5V output voltage from a higher input voltage. It is often used to power 5V logic circuits, microcontrollers, and other digital components.

Key Features

- 1. Output Voltage:**
 - **Fixed Output:** Provides a constant 5V output voltage.
- 2. Input Voltage Range:**
 - **Typical Input Range:** 7V to 12V (depending on the specific variant and load conditions). The input voltage must always be higher than the output voltage by at least a few volts to maintain regulation.
- 3. Current Capacity:**
 - **Maximum Output Current:** Typically, up to 1A. Some versions may handle slightly more, but it's best to refer to the datasheet for specific current ratings.
- 4. Pin Configuration:**
 - **Pin 1 (Input):** Connects to the higher voltage supply (input voltage).
 - **Pin 2 (Ground):** Connects to the ground of the circuit.
 - **Pin 3 (Output):** Provides the regulated 5V output voltage.

5. Thermal Protection:

- Includes thermal overload protection, which shuts down the regulator if it overheats, preventing damage.

6. Short-Circuit Protection:

- Equipped with short-circuit protection to prevent damage if the output is shorted.

3.1.7 BREAD BOARD



Fig.3.7 Bread board

A breadboard is a fundamental tool in electronics for building and testing circuits without soldering. It features a grid of interconnected holes where electronic components and wires can be inserted to create temporary circuits. Breadboards are divided into two main sections: terminal strips and bus strips.

3.1.7.1 Terminal Strips: The central area where most components are placed, with rows of five connected holes.

3.1.7.2 Bus Strips: Located on the sides, used for distributing power and ground, typically marked with red (positive) and blue (negative) lines.

Usage:

3.1.7.3 Insert Components: Place components into the holes, ensuring correct orientation for polarized parts.

3.1.7.4 Connect Wires: Use jumper wires to link different parts of the circuit.

3.1.7.5 Power Distribution: Connect power and ground to the bus strips from a power supply.

3.1.7.6 Test and Modify: Power the circuit, test its functionality, and easily modify as needed.

Benefits:

3.1.7.7 No Soldering: Allows easy changes and adjustments.

3.1.7.8 Reusable: Components can be removed and reused.

3.1.7.9 User-Friendly: Suitable for beginners and professionals.

3.1.7.10 Versatile: Supports a variety of components.

3.1.8 JUMPER WIRES



Fig.3.8 Jumper Wires

Jumper wires are flexible electrical cables used to establish temporary or permanent connections between different parts of an electronic circuit. They are particularly useful for prototyping and testing circuits on breadboards and other development platforms.

Key Features

1. Construction:

- **Wire Material:** Typically made from stranded copper wire, which provides flexibility and durability.
- **Insulation:** Covered with a plastic or rubber coating to prevent short circuits and to protect the wire.

2. Connectors Types:

- **Male-to-Male:** These jumper wires have male pins at both ends, designed for insertion into female headers or breadboard holes.
- **Female-to-Female:** Equipped with female sockets on both ends, suitable for connecting to male pins or headers.
- **Male-to-Female:** One end has a male pin and the other end has a female socket, allowing it to connect a male pin to a female header or vice versa.
- **Compatibility:** Connectors are often standardized to fit typical breadboard and pin header sizes (e.g., 2.54 mm pitch).

3. Colours:

- **Color Coding:** Jumper wires come in various colors, such as red, black, yellow, green, and blue. This color-coding helps in organizing and identifying different connections in a circuit, improving clarity and reducing errors.

4. Length and Flexibility:

- **Lengths:** Available in various lengths, commonly ranging from a few centimetres to several inches, allowing for flexibility in circuit design.
- **Flexibility:** The wire is typically flexible, which facilitates easy routing and adjustment of connections on a breadboard.

3.1.9 9v dc supply cell



Fig.3.9 Jumper Wires

A 9V DC supply cell, often referred to as a 9V battery, is a common type of battery used in various electronic devices. It provides a constant voltage of 9 volts and is widely used in applications such as smoke detectors, small electronic projects, and some radios. A 9V battery is a compact battery that delivers a nominal voltage of 9 volts, widely used in various electronic devices. It comes in several types, including alkaline, rechargeable (NiMH and lithium-ion), and lithium. Alkaline batteries are common for everyday use in devices like smoke detectors and remote controls due to their affordability and reasonable capacity of around 500-600 mAh. Rechargeable 9V batteries, such as NiMH, are more economical over time and suitable for frequent use, though they generally offer lower capacity compared to alkaline types. Lithium 9V batteries provide a higher capacity and longer shelf life, making them ideal for critical applications where reliability is essential. 9V batteries are versatile and found in devices such as musical equipment, handheld meters, and certain radios. They are valued for their ability to deliver consistent power. However, users should handle them with care to avoid short circuits and dispose of them according to local regulations due to the potential presence of hazardous materials. Proper storage in a cool, dry place can extend their shelf life.

3.2 SOFTWARE DESCRIPTION

3.2.1 ARDUINO IDE



Fig.3.10 Arduino IDE

An Integrated Development Environment (IDE) is a comprehensive software application that facilitates software development by providing tools and features in a unified interface. IDEs

typically include a code editor, a compiler or interpreter, a debugger, and build automation tools. The primary purpose of an IDE is to streamline the development process, enhance productivity, and simplify debugging.

One key advantage of using an IDE is the integration of multiple development tools, which allows developers to write, test, and debug code within a single environment. This integration reduces the need to switch between different applications and helps maintain workflow efficiency. IDEs often come with features like syntax highlighting, code completion, and error detection, which assist in writing code more quickly and accurately.

3.2.2 TINKER-CAD SIMULATOR



Fig.3.11 Tinker-CAD simulator

"Tinker cad Simulator is a powerful tool designed for electronics enthusiasts, hobbyists, and students to simulate circuits and experiment with electronics virtually. Developed by Autodesk, it provides a user-friendly interface that allows users to drag and drop components such as resistors, capacitors, transistors, and more onto a virtual breadboard.

One of its key features is its ability to simulate the behavior of circuits in real-time, helping users understand how components interact without needing physical hardware. This virtual environment is invaluable for learning electronics principles, troubleshooting circuits, and experimenting with designs before moving to physical prototypes.

Tinker cad Simulator also supports Arduino simulation, enabling users to program and simulate Arduino microcontroller boards within the same platform. This integration is particularly beneficial for IoT (Internet of Things) projects, robotics, and embedded systems development

CHAPTER 4

SYSTEM AND DESIGN DESCRIPTION

4.1 BLOCK DIAGRAM

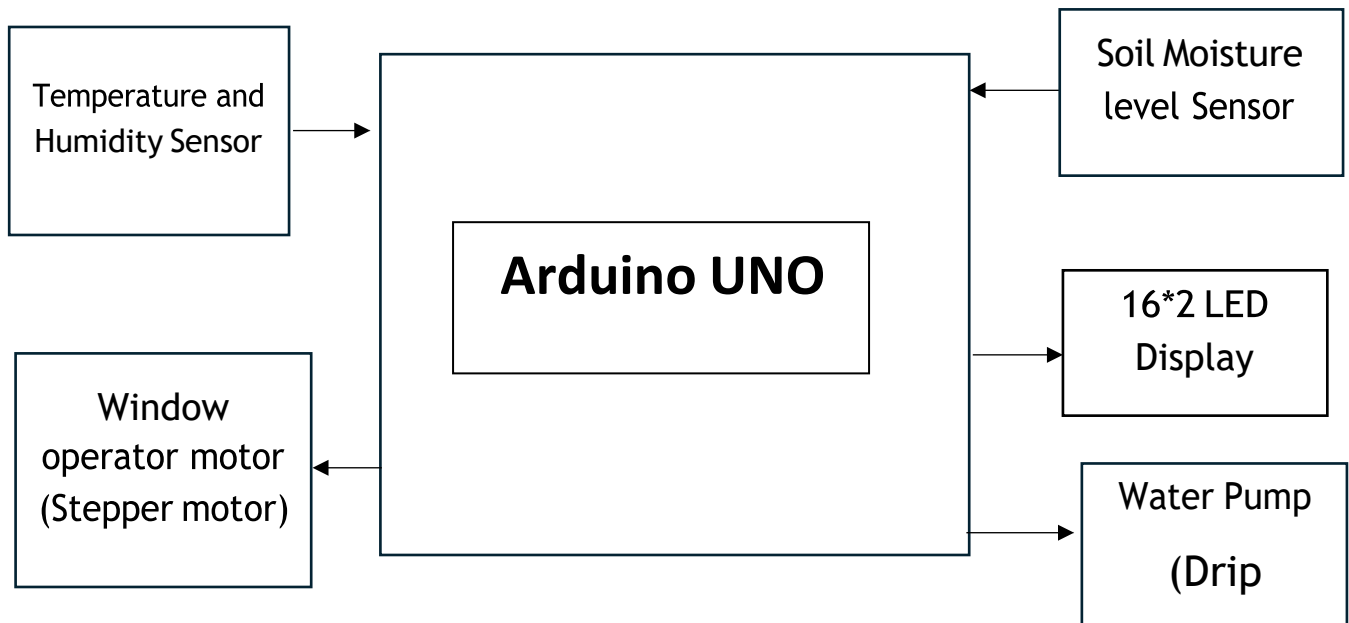


Fig.4.1 Real Time Responsive Automated Window and Water Supply Operator for Indoor Plantations using IOT

Fig.3.1 represents the block diagram presented the interfacing of various critical components with an Arduino UNO board, which serves as the central control unit for Real Time Responsive Automated Window and Water Supply Operator for Indoor Plantations using IOT.

The Arduino UNO plays the major role like connecting all the input and output components and the main and important thing is we can store the code in the Arduino UNO board it will use full to perform the operations.

4.2 CIRCUIT DIAGRAM

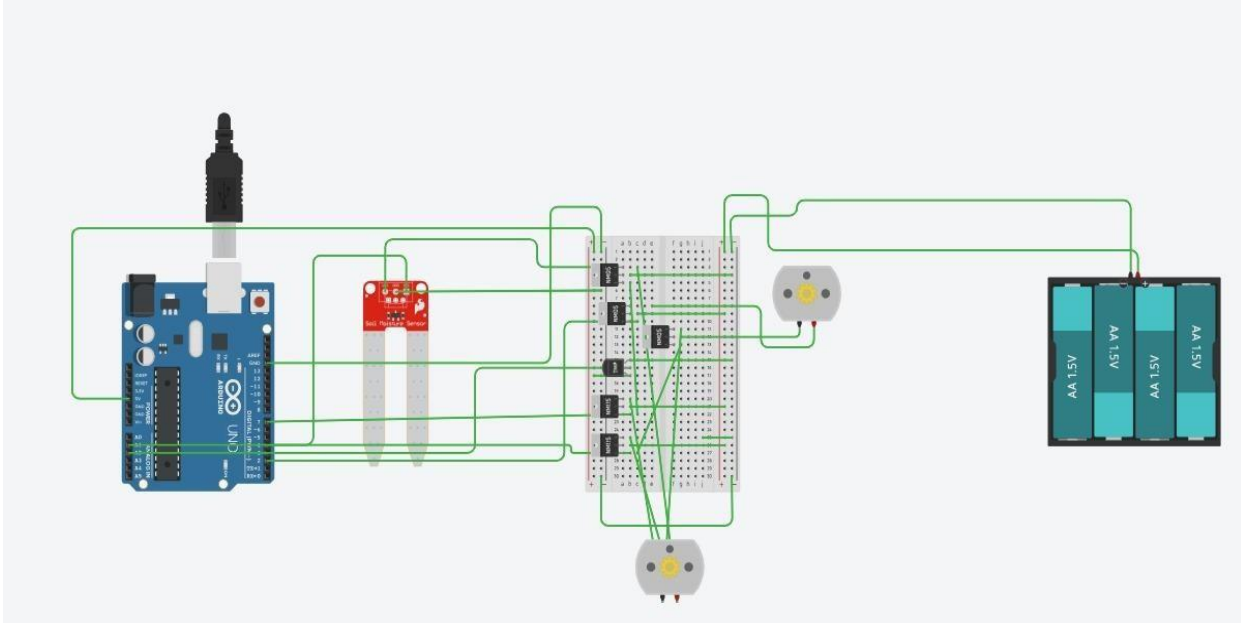


Fig.4.2 Circuit Diagram

COMPONENT COLLECTION AND PREPARATION

- ☐ Arduino UNO:
- ☐ Soil Moisture sensor:
- ☐ Water pump:
- ☐ DHT11 sensor: The sensor is used to detect the temperature and humidity
- ☐ Stepper motor and driver:
- ☐ Bread Board: Ic's placement and all wiring are done on the board.
- ☐ Jumper Wires:

CIRCUIT ASSEMBLY

- ☐ Follow the provided circuit diagram to connect the Arduino UNO, soil moisture sensor ,water pump,dht11 sensor ,stepper motor and driver, jumper wire.
- ☐ Pay attention to polarity and connections to avoid short circuits and potential damage.

SOFTWARE SETUP

☐ Arduino IDE: Install the Arduino IDE and load the necessary firmware (e.g., Marlin) onto the Arduino Mega.

☐ Tinker-cad simulator

CALIBRATION AND TESTING

☐ Endstop Configuration: Ensure endstops are correctly positioned and configured for accurate homing of the print head.

☐ Motor Calibration: Calibrate the stepper motors to ensure precise positioned.

☐ Temperature Calibration: Adjust the heater and temperature sensor settings to maintain the optimal temperature.

☐ Test Prints: Conduct initial test to verify the functionality and accuracy to Make necessary adjustments based on the results.

4.3 PROGRAM CODE

```
#include "DHT.h"

#define DHTPIN 4    // Digital pin connected to the DHT sensor

#define DHTTYPE DHT11 // DHT 11

DHT dht(DHTPIN, DHTTYPE);

int count=0;

#define m1 2

int s;

void setup()
{
    Serial.begin(9600);
    pinMode(m1, OUTPUT);
    dht.begin();
}

void loop()
{
    delay(900);
    s = analogRead(A5);
    float h = dht.readHumidity();
    float t = dht.readTemperature();
    Serial.println(h);
    Serial.println(s);
    if(s>600)
    {
        digitalWrite(m1, 1);
    }
    else{
        digitalWrite(m1, 0);
    }
}
```



```
if(h>40 && count!=1)
{
    step(1);
    delay(500);
    count++;
}
else if(h<=40 && count!=0)
{
    step(-1);
    delay(500);
    count--;
}
}

void step(int a)
{
    int arr[]={1,0,1,1};
    if(a>0){
        for(int i=0;i<444;i++)
        {
            digitalWrite(7,arr[0]);
            digitalWrite(8,arr[1]);
            digitalWrite(12,arr[2]);
            digitalWrite(13,arr[3]);
            int temp=arr[3];
            arr[3]=arr[2];
            arr[2]=arr[1];
            arr[1]=arr[0];
            arr[0]=temp;
            delay(5);
        }
    }
}
```

```
}  
else  
{  
  for(int i=0;i<444;i++)  
  {  
    digitalWrite(7,arr[0]);  
    digitalWrite(8,arr[1]);  
    digitalWrite(12,arr[2]);  
    digitalWrite(13,arr[3]);  
    int temp=arr[0];  
    arr[0]=arr[1];  
    arr[1]=arr[2];  
    arr[2]=arr[3];  
    arr[3]=temp;  
    delay(5);  
  }  
}  
}
```

4.4 ADVANTAGES, APPLICATIONS AND LIMITATIONS

Advantages:

- Saves manual monitoring time.
- Ensures atmospheric parameter's quality.
- No extra labor cost required for the operation.
- Safe protection from weather disruptions such as heavy rain, storms, strong winds etc.

Applications :

- The project targets indoor farms and plantation nurseries .
- Targeted specially on indoor plantation farms that are located in remote areas or in countries where water scarcity is serious factor.
- This set-up of the project can be alternatively modified to captivate endangered species of animals such as rare apes, birds, fish that are breed in captivity of specific temperature habitats.

Limitations :

- Cost of maintenance can be high cause of faulty components.
- Normal gardener / Normal Zoo-keeper cannot operate the system without specific training/certified knowledge about IOT.
- Margin for accuracy errors and risk factors of system fail is always to be considered.

CONCLUSION

The project "Real Time Responsive Automated Window and Water Supply Operator for Indoor Plantations using IoT" has successfully demonstrated the feasibility and effectiveness of integrating IoT technologies into indoor farming environments. Throughout the development and implementation phases, several key achievements and findings have emerged:

1. **Automation and Efficiency:** By automating the control of windows and water supply based on real-time environmental data (such as, soil moisture, and temperature), the system has significantly improved operational efficiency in indoor plantations. This automation reduces manual intervention, optimizes resource utilization, and enhances crop health and growth.
2. **Reliability and Responsiveness:** Through rigorous testing and validation, the system has proven to be reliable and responsive under various environmental conditions. It effectively adjusts window openings and water supply mechanisms in real-time, ensuring that plants receive optimal conditions for growth.
3. **User Interface and Interaction:** The development of a user-friendly interface has empowered users to monitor and manage the indoor plantation environment effortlessly. Features such as remote monitoring, setting preferences, and receiving alerts have enhanced user experience and control over the system.
4. **Challenges and Solutions:** Challenges encountered during the project, including hardware limitations and connectivity issues, were systematically addressed. Solutions implemented have improved system robustness and performance.

FUTURE SCOPE

1. **Advanced Sensor Integration:** Incorporating advanced sensors for more precise environmental monitoring (e.g., CO2 levels, nutrient concentrations) can further optimize plant growth conditions.
2. **Predictive Analytics and Machine Learning:** Implementing predictive analytics and machine learning algorithms can anticipate environmental changes and optimize system responses proactively.
3. **Energy Efficiency:** Exploring energy-efficient technologies (such as solar-powered components or energy harvesting) can reduce operational costs and environmental impact.
4. **Scalability and Adaptability:** Scaling the system for larger indoor plantations or adapting it to different crops and environments will broaden its applicability and impact.
5. **IoT Platform Enhancements:** Continuously updating and integrating with advanced IoT platforms to leverage new features and capabilities in data analytics and management.
6. **Community and Industry Collaboration:** Collaborating with indoor farming communities and industry stakeholders can foster innovation and adoption of IoT-driven solutions in agriculture.

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